ECON 0100 | Vignette D2 | Commons

Solutions

Answers and work are shown in the red blocks.

Due in Recitation. Vignettes are a certificate of the work done together in Recitation.

Late Night Ingredients

Two skilled potion makers, Slughorn and Snape, can each brew different types of potions. They share access to a garden where they gather rare ingredients for their potions. The ingredients can only be harvested late at night, making it difficult to know who else uses the garden. Each must decide how much of the ingredients to harvest. The payoff matrix shows their monthly profits in Coins:

		Snape	
		S	OH
Slughorn	Sustainable (S)	8, 8	2, 12
	Over-Harvest (OH)	12, 2	4, 4

No solution notes exist for this question. These answers are Claude's, worked from the payoff matrix.

a) What is Slughorn's best response when Snape chooses Sustainable? _____

Over-Harvest: $12 > 8$.

b) What is Slughorn's best response when Snape chooses Over-harvest? _____

Over-Harvest: $4 > 2$.

c) What is *(are)* the Nash Equilibrium? _____

(OH, OH). Over-harvesting is each brewer's best response whatever the other does (the game is symmetric), so both over-harvest.

d) What is the socially efficient outcome? _____

(S, S). Total monthly profits are 16 at (S, S), 14 at (S, OH) and (OH, S), and 8 at (OH, OH).

Best responses are boxed. The yellow cell is the Nash equilibrium, and the blue cell is socially efficient.

		Snape	
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